

Algorithm Analysis And Data Structures

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2.4 Running Time Calculation

Example: Maximum Subsequence Sum Problem

$A[] = \{4, -3, 5, 2, -1, 2, 6, -2\}$

$\text{Max_Sum} = ?$

```

int max_subsequence_sum( int a[], unsigned int n )
{
    int this_sum, max_sum, best_i, best_j, i, j, k;

    max_sum = 0; best_i = best_j = -1;
    for( i=0; i<n; i++ )
        for( j=i; j<n; j++ )
            {
                this_sum=0;

                for( k = i; k<=j; k++ )
                    this_sum += a[k];
                if( this_sum > max_sum )
                {
                    max_sum = this_sum;
                    best_i = i;
                    best_j = j;
                }
            }
    return( max_sum );
}

```

$O(N^3)$

Figure 2.5 Algorithm 1

```

int max_subsequence_sum( int a[], unsigned int n )
{
    int this_sum, max_sum, best_i, best_j, i, j, k;

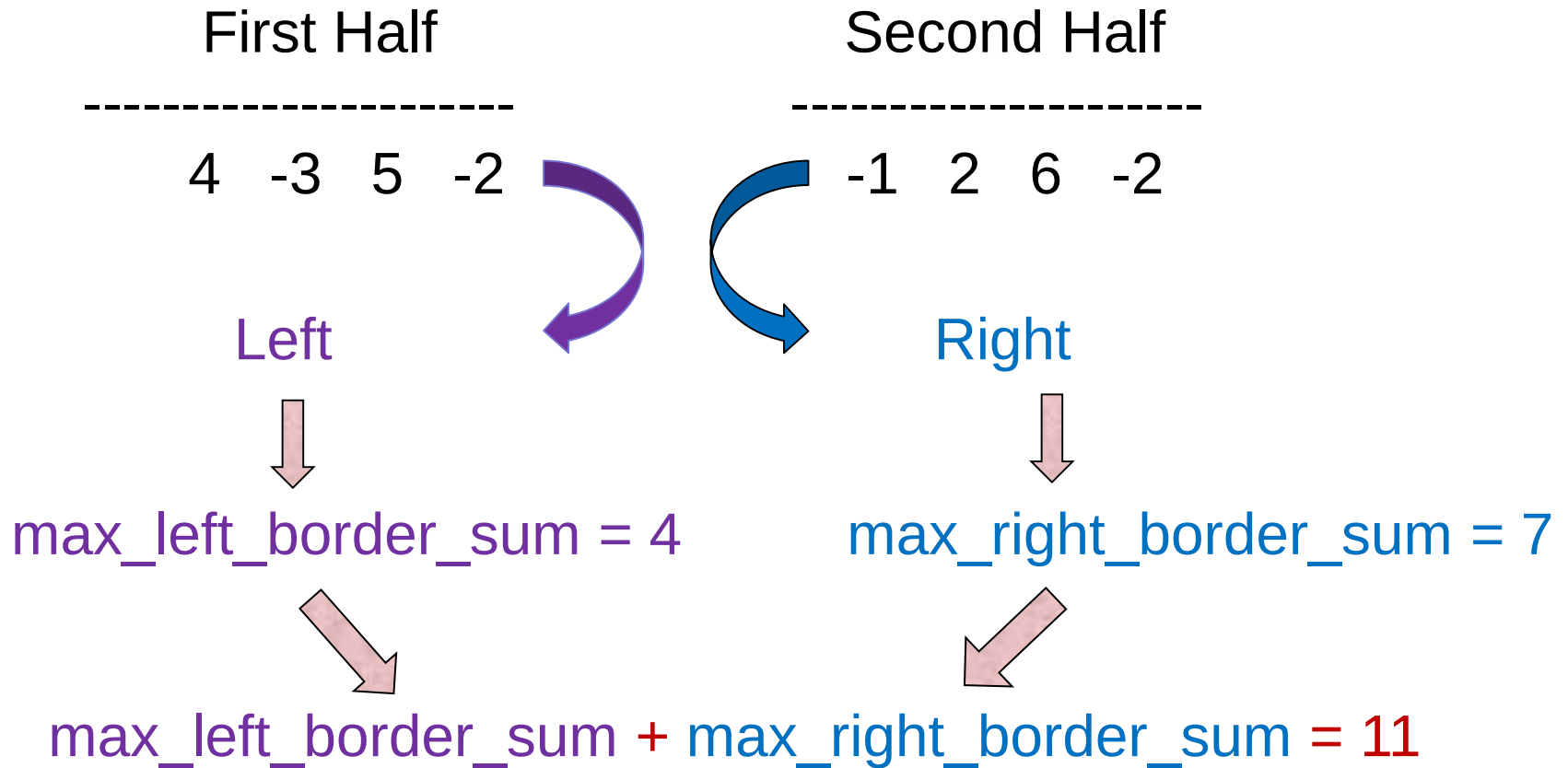
    max_sum = 0; best_i = best_j = -1;
    for( i=0; i<n; i++ )
    {
        this_sum = 0;
        for( j=i; j<n; j++ )
        {
            this_sum += a[j];
            if( this_sum > max_sum )
            {
                max_sum = this_sum;
                best_i = i;
                best_j = j;
            }
        }
    }
    return( max_sum );
}

```

$O(N^2)$

Figure 2.6 Algorithm 2

Divide&Conquer Algorithm



```
int max_sub_sequence_sum( int a[], unsigned int n )
{
    return max_sub_sum( a, 0, n-1 );
}
```

```
int max_sub_sum( int a[], int left, int right )
{
    int max_left_sum, max_right_sum;
    int max_left_border_sum, max_right_border_sum;
    int left_border_sum, right_border_sum;
    int center, i;
    if ( left == right )
        if( a[left] > 0 ) return a[left];
    else return 0;

    center = (left + right )/2;
    max_left_sum = max_sub_sum( a, left, center );
    max_right_sum = max_sub_sum( a, center+1, right );

    max_left_border_sum = 0; left_border_sum = 0;
    for( i=center; i>=left; i-- )
    {
        left_border_sum += a[i];
        if( left_border_sum > max_left_border_sum )
            max_left_border_sum = left_border_sum;
    }

    max_right_border_sum = 0; right_border_sum = 0;
    for( i=center+1; i<=right; i++ )
    {
        right_border_sum += a[i];
        if( right_border_sum > max_right_border_sum )
            max_right_border_sum = right_border_sum;
    }

    return max3( max_left_sum, max_right_sum,
                max_left_border_sum + max_right_border_sum );
}
```

$O(N \cdot \log N)$

Figure 2.7 Algorithm 3 - Divide&Conquer Algorithm

```

int max_subsequence_sum( int a[], unsigned int n )
{
    int this_sum, max_sum, best_i, best_j, i, j;
    i = this_sum = max_sum = 0; best_i = best_j = -1;
    for( j=0; j<n; j++ )
    {
        this_sum += a[j];
        if( this_sum > max_sum )
        {
            max_sum = this_sum;
            best_i = i;
            best_j = j;
        }
        else
        if( this_sum < 0 )
        {
            i = j + 1;
            this_sum = 0;
        }
    }
    return( max_sum );
}

```

$O(N)$

Figure 2.8 Algorithm 4

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Questions and Answers

Thank you ...

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